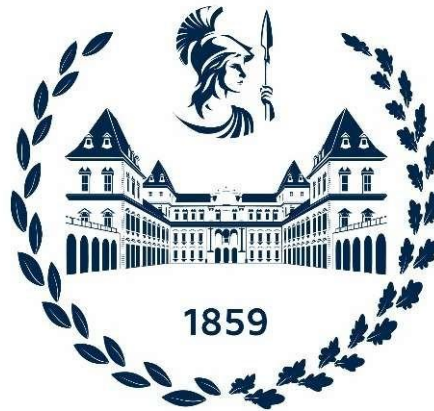


BHT examples

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Exe #1, Point #1

Let consider the Branch Prediction Mechanism based on the Branch History Table (BHT).

You are requested to

1. Describe the architecture of a BHT

The BHT is a table composed of N elements of M bits each (typically, 1 or 2).

Exe #1, Point #2

- 2. Describe in details the behavior of a BHT: when it is accessed, which input and output information are involved with each access*

Each time a processor decodes a conditional branch instruction, it accesses the BHT using the lowest bits of its address (ignoring those related to the instruction alignment). The selected BHT entry tells whether the instruction should be predicted Taken or NotTaken, based on previous executions of the same instruction. When the results of the branch is known, the BHT is possibly updated.

Exe #1, Point #3

- 3. Assuming that the processor uses 32 bit addresses, each instruction is 4 byte wide, and the BHT is composed of 16 entries, clarify the content and size of the fields composing each BHT entry.*

The BHT in this case is composed of 16 entries. Bits 0 and 1 are ignored for indexing, since each instruction takes 4 bytes. Bits 2:5 are used for indexing. Each row contains an n-bit saturating counter.

Exe #1, Point #4

4. Assuming that the processor uses 32 bit addresses, each instruction is 4 byte wide, and the BHT is composed of 16 entries, identify the final content of the BHT if
- The BHT is initially empty (i.e., full of 0s, meaning NotTaken)
 - The following instructions are executed in sequence
 - l.d r2,v1(r1) located at the address 0x00A50050
 - bnez r2,l1 located at the address 0x00A50054; the branch is taken, and the branch target address is 0x00A60050
 - addi r2, r2, #1 located at the address 0x00A60050
 - bez r2,l2 located at the address 0x00A60054; the branch is not taken
 - andi r5, r5, #1 located at the address 0x00A60058
 - bez r2,l3 located at the address 0x00A6005C; the branch is taken, and the branch target address is 0x00BB0040
 - add r1, r2, r3 located at the address 0x00BB0040
 - bez r1,l4 located at the address 0x00BB0044; the branch is taken, and the branch target address is 0x00A50050.

Exe #1, Point #4: BHT content after l.d execution

I.d address	
0x00A50050	0000 0000 1010 0101 0000 0000 0101 0000

$$\text{l.d } r2, v1(r1)$$
[illegible]

Exe #1, Point #4: BHT content after bnez execution

branch taken

bnez address	
0x00A50054	0000 0000 1010 0101 0000 0000 0101 0100

bnez r2,l1

the branch is taken, and the branch target address is 0x00A60050

0
0
0
0
0
1
0
0
0
0
0
0
0
0
0
0

Entry #5 is accessed.
The entry value was 0
(Predict NotTaken) →
Misprediction

Exe #1, Point #4: BHT content after addi execution

addi address	
0x00A60050	0000 0000 1010 0101 0000 0000 0101 0000

```
addi r2, r2, #1
```

[illegible]

Exe #1, Point #4: BHT content after bez execution

branch not taken

bez address	
0x00A60054	0000 0000 1010 0110 0000 0000 01 01 0100

bez r2,l2

0
0
0
0
0
0
0
0
0
0
0
0
0
0
0
0

Entry #5 is accessed.
The entry value was 1
(Predict Taken) →
Misprediction

Exe #1, Point #4: BHT content after andi execution

	andi address
0x00A60058	0000 0000 1010 0101 0000 0000 0101 1000

```
andi r5, r5, #1
```

[illegible]

Exe #1, Point #4: BHT content after bez execution

branch taken

bez address	
0x00A6005C	0000 0000 1010 0101 0000 0000 0101 1100

bez r2,l3

the branch is taken, and the branch
target address is 0x00BB0040

0
0
0
0
0
0
0
1
0
0
0
0
0
0
0
0

Entry #7 is accessed.
The entry value was 0
(Predict NotTaken) →
Misprediction

Exe #1, Point #4: BHT content after add execution

add address	
0x00BB0040	0000 0000 1011 1011 0000 0000 0100 0000

```
add r1, r2, r3
```

[illegible]

Exe #1, Point #4: BHT content after bez execution

branch taken

bez address	
0x00BB0044	0000 0000 1011 1011 0000 0000 0100 0100

bez r1,l4

the branch is taken, and the branch
target address is 0x00A50050

0
1
0
0
0
0
0
1
0
0
0
0
0
0
0
0

Entry #1 is accessed.
The entry value was 0
(Predict NotTaken) →
Misprediction